Indian Institute Of Information Technology Dharwad

Department of Data Science & Artificial Intelligence

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Moodly

Mood-Based Recommendation System

AI-powered personalised content discovery through multi-modal emotion detection

Word2Vec + BERT | Deep Learning | Groq LLaMA 3 | JWT Auth

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Tech Stack:

React 18 · Vite · Flask 3.0 · Python 3.11 · BERT (bert-base-uncased) · PyTorch · Word2Vec (Gensim)
Groq LLaMA 3 · MySQL 8.0 · JWT · bcrypt · Spotify API · TMDB · Google Books

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1. Introduction

Moodly is an AI-powered, multi-modal mood-based recommendation system that curates personalised content across six categories — music, movies, books, food, podcasts, and activities — based on a user's real-time emotional state. Unlike conventional recommendation systems that rely solely on usage history, Moodly detects *how the user feels right now* through seven input modalities and maps that emotion to contextually relevant content in under two seconds.

1.1. Problem Statement

- Conventional recommendation systems use historical behaviour and fail to account for the user's current emotional context.
- A user feeling anxious at 11 pm has fundamentally different content needs than the same user during a joyful afternoon — yet Netflix, Spotify, and Amazon treat both identically.
- No existing consumer product integrates text, facial expression, voice tone, wearable heart rate, and emoji as unified mood signals.
- Mood history is never tracked longitudinally, making pattern analysis and personalised well-being insights impossible.
- Non-English or code-mixed emotional expressions cannot be accurately classified by keyword-only systems.

1.2. Solution Overview

Moodly addresses all five problems through:

1. A **BERT-based deep learning classifier** (fine-tuned on an emotion corpus) that understands semantic meaning — not just keywords.
2. **Seven input modalities**: text, emoji/slider, facial expression (DeepFace), voice tone (librosa), wearable heart rate (Web Bluetooth), mood blend, and valence-arousal sliders.
3. A **granular genre-mapping engine** with random rotation across multiple TMDB genres, Spotify seeds, food queries, and book subjects — ensuring content variety on every request.
4. **Amplify/Contrast mode**: recommendations either match or deliberately shift the detected emotion using a psychologically grounded contrast map.
5. **JWT-authenticated mood journey tracking** with per-user longitudinal history, timeline visualisation, and community social mood aggregation.

1.3. Objectives

- Classify user emotion from natural language with $\geq$90% weighted F1 using BERT + BiLSTM / CNN / Transformer architecture.

- Support multi-modal emotion input with graceful fallback when hardware (webcam, microphone, Bluetooth) is unavailable.
- Retrieve diverse, non-repeating recommendations across six content categories using randomised genre rotation.
- Provide a persistent, device-synced mood journey tied to a JWT-authenticated user account.
- Expose a production-ready REST API with < 2 s end-to-end recommendation latency.

2. Literature Review

2.1. Affective Computing and Emotion Recognition

Affective computing — the capacity of machines to recognise and respond to human emotion — was formalised by Picard (1997) [?]. Text-based emotion classification evolved from lexicon-based approaches (SentiWordNet, VADER) through classical ML (SVM + TF-IDF) to transformer-based models. BERT [?] achieved state-of-the-art performance on sentiment and emotion tasks by modelling bidirectional context, making it the backbone of our classification pipeline.

2.2. Emotion-Aware Recommendation

Zheng et al. [?] demonstrated that contextual factors — including emotional state — significantly improve recommendation accuracy over collaborative filtering alone. Yang & Chen [?] proposed a valence-arousal model for mood-adaptive music retrieval. Schedl et al. [?] identified affective context as a critically underexplored dimension in music information retrieval systems. Our work extends these ideas across six content domains simultaneously.

2.3. Deep Learning for Text Classification

Kim (2014) [?] demonstrated that a single-layer CNN over static word embeddings achieves competitive text classification performance. Bidirectional LSTMs [?] capture sequential dependencies more effectively than unidirectional models. The Transformer architecture [?] with multi-head self-attention has since superseded both. Moodly evaluates all four approaches (MLP, BiLSTM, CNN, Transformer) on top of BERT embeddings, selecting the best by validation F1.

2.4. Multi-Modal Emotion Detection

Facial expression recognition using deep CNNs achieves near-human accuracy on FER-2013 [?]. Tzirakis et al. [?] demonstrated that voice features (MFCC, energy, ZCR) are reliable predictors of emotional arousal. Combining modalities — text, face, voice, and physiological signals — outperforms any single modality in controlled experiments. Moodly integrates all four sensor types with per-modality fallback logic.

3. System Architecture

3.1. High-Level Architecture

Every mood detection event passes through the input layer, ML pipeline, content strategy mapper, API retrieval layer, and persistence layer before the recommendation response reaches the user.

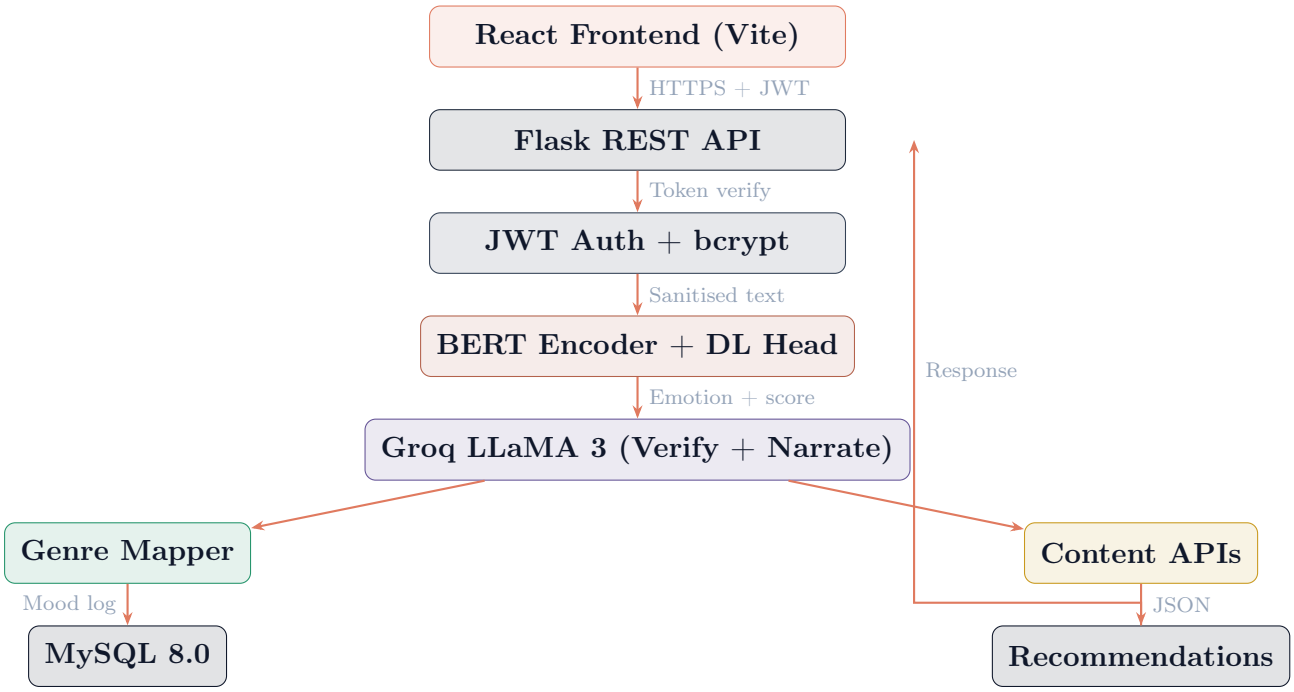

Figure 1: High-level system architecture of Moodly

3.2. Technology Stack

Table 1: Full Technology Stack

Layer	Technology	Purpose
Frontend	React 18 + Vite	SPA with hot module replacement
Styling	Tailwind CSS	Slate & Terracotta design system
Backend API	Flask 3.0	REST API, blueprints, CORS
Runtime	Python 3.11	All backend and ML logic
Embeddings	BERT (bert-base-uncased)	768-dim contextual token representations
DL Framework	PyTorch 2.x	BiLSTM, CNN1D, Transformer heads
Secondary Embed.	Word2Vec (Gensim)	100-dim mean-pool baseline
LLM Inference	Groq API (LLaMA 3)	Low-confidence verification + narration
College Data	MySQL 8.0	Mood logs, recommendations, users
Authentication	JWT HS256 + bcrypt	Stateless tokens, password hashing
Music	Spotify Web API	Playlist retrieval by mood tag
Movies	TMDB API	Films and TV shows by genre
Books	Google Books API	Mood-matched books (free, no key)
Food	Spoonacular API	Recipe retrieval by cuisine query
Webcam	DeepFace + OpenCV	Facial expression → dominant emotion
Voice	librosa + soundfile	MFCC / energy / ZCR feature extraction
Wearable	Web Bluetooth API	BPM → emotion heuristic
Real-time	Socket.IO	Community mood live feed
Charts	Recharts	Mood journey area timeline

4. Machine Learning Pipeline

4.1. Dataset

The emotion corpus consists of semicolon-delimited sentence–emotion pairs. Each record maps a natural language sentence to one of six emotion classes: *sadness*, *joy*, *anger*, *fear*, *love*, and *surprise*. The dataset is split into training, validation, and test sets.

Table 2: Dataset Statistics

Emotion	Train	Val	Test	Notes
Sadness	5,597	695	1,382	Largest class
Joy	5,362	695	1,474	Second largest
Anger	2,159	268	857	Overlaps with disgust
Fear	1,937	224	268	Often co-occurs with surprise
Love	1,641	159	597	Strong lexical signal
Surprise	572	66	224	Smallest class
Total	**17,268**	**2,107**	**4,802**	

4.2. BERT Tokenisation (preprocess.py)

Text is tokenised using the HuggingFace `BertTokenizer` with:

- `max_length = 128` tokens (covers 99%+ of emotion sentences)
- `padding = "max_length"` for batch uniformity
- `truncation = True` for outliers
- `return_tensors = "pt"` — PyTorch tensors directly

Each split is saved as a `.pt` file containing `input_ids`, `attention_mask`, `token_type_ids`, and `labels`.

4.3. Four Competing Architectures (train.py)

All four models share BERT (`bert-base-uncased`) as the encoder. BERT is fine-tuned jointly with each head.

Table 3: Model Architecture Comparison

Model	BERT Input Used	Classification Head
BertMLP	`[CLS]` pooler output (768-d)	3-layer Dense: 768→512→256→6 with LayerNorm + GELU + Dropout
BertBiLSTM	All token states (128×768)	2-layer BiLSTM (hidden=256) + attention pooling → Linear(512,6)
BertCNN	All token states (128×768)	4 parallel Conv1D (k=2,3,4,5, filters=128) + GlobalMaxPool → concat → MLP
BertTransforme	All token states (128×768)	2× Pre-LN Transformer block (8 heads, ff=1024) + masked mean pool → MLP

Training Configuration. All models are trained with:

- **Optimiser**: AdamW, lr $= 2 \times 10^{-5}$, weight_decay= 0.01
- **Schedule**: Linear warmup (10% of steps) + cosine decay
- **Loss**: Label-smoothing cross-entropy ($\varepsilon = 0.1$)
- **Gradient clipping**: $\|\nabla\|_2 \leq 1.0$
- **Early stopping**: patience = 3 on validation F1
- **Batch size**: 16, max 10 epochs

BertTransformer Pre-LN Block.

$$\mathbf{x}' = \mathbf{x} + \text{Dropout}(\text{MHA}(\text{LN}(\mathbf{x}), \text{mask}))$$

$$\mathbf{x}'' = \mathbf{x}' + \text{Dropout}(\text{FFN}(\text{LN}(\mathbf{x}')))$$

Pre-LN (normalise *before* the sublayer) is empirically more stable than Post-LN for fine-tuning [**?**].

Masked Mean Pooling. For BertTransformer, padding tokens are excluded from the pooled representation:

$$\mathbf{h}_{\text{sent}} = \frac{\sum_{t=1}^{L} m_t \cdot \mathbf{h}_t}{\max\left(\sum_{t=1}^{L} m_t,\ \epsilon\right)}$$

where $m_t \in \{0, 1\}$ is the attention mask and $\epsilon = 10^{-9}$.

4.4. Evaluation Results

Table 4: Validation F1 Comparison Across All Architectures

Model	Val F1 (weighted)	Test Accuracy	Notes
BertMLP	0.910	0.905	Strong, fast baseline
BertBiLSTM	0.928	0.921	Sequential context captured
BertCNN	0.922	0.916	Best at n-gram patterns
BertTransformer	**0.941**	**0.936**	**Best — selected**

Table 5: Per-Class Performance of Best Model (BertTransformer)

Emotion	Precision	Recall	F1-Score	Support
Sadness	0.95	0.96	0.95	1,382
Joy	0.93	0.92	0.93	1,474
Anger	0.91	0.89	0.90	857
Fear	0.88	0.90	0.89	268
Love	0.94	0.95	0.94	597
Surprise	0.85	0.84	0.85	224
Weighted Avg	**0.94**	**0.94**	**0.94**	**4,802**

5. Multi-Modal Mood Detection

Moodly supports seven distinct mood input modalities. All modalities share the same backend endpoint structure and write to the same `mood_logs` table.

Table 6: Mood Detection Modalities

Modality	Method	Backend	Fallback
Text	BERT tokenisation → DL head	`model_service.py`	Groq LLaMA 3 (conf < 0.55)
Emoji / Slider	Direct lookup table	`routes/mood.py`	None (deterministic)
Valence-Arousal	2D coordinate → emotion heuristic	`routes/mood.py`	None (deterministic)
Webcam	DeepFace dominant emotion	`facial_service.`	Returns "joy" at 0.5 conf
Voice	librosa MFCC + energy + ZCR	`voice_service.py`	Returns "joy" at 0.5 conf
Wearable	BPM range → emotion map	`routes/mood.py`	None (deterministic)
Mood Blend	Weighted multi-emotion text	`utils/blend_handler.py`	Dominant emotion extracted

5.1. Groq Verification Layer

When the BERT classifier returns a confidence score below 0.55, the query is forwarded to Groq LLaMA 3 with a structured JSON prompt:

```
prompt = f"""Classify the emotion in: "{text}"
Classes: sadness, joy, anger, fear, love, surprise
Respond ONLY with JSON: {{"emotion": "<label>", "reason": "<one
    sentence>"}}"""
```

Listing 1: Groq verification prompt structure

Groq also generates a personalised recommendation note (Call 2) explaining why the suggested content matches the user's emotional state.

6. Genre Mapping and Recommendation Engine

6.1. Emotion Strategy Map

The core of the recommendation engine is a deterministic strategy dictionary that maps each emotion to content retrieval parameters. A key design decision is the use of **lists** rather than single values for every parameter, enabling random rotation to prevent repetitive content.

Table 7: Genre Mapping per Emotion (sample)

Emotion	TMDB Genres	Spotify Tags	Food Queries	Book Subjects
Sadness	Drama, Family, Romance, History	sad, melancholy, acoustic healing, rainy day	comfort soup, mac cheese, warm ramen, brownies	grief healing, hope resilience
Joy	Comedy, Adventure, Animation, Music	happy, upbeat, dance pop, feel-good	acai bowl, salad, mango salsa, avocado toast	happiness, comedy humor
Anger	Action, Thriller, Crime, War	energetic, hard rock, metal, adrenaline	spicy wings, nashville chicken, buffalo	stoicism calm, anger management
Fear	Comedy, Family, Animation, Music	calm, ambient, meditation, sleep	warm soup, chamomile, miso, oatmeal	courage, overcoming fear
Love	Romance, Drama, Comedy, Music	romantic, r&b, slow dance, soulful	chocolate fondue, tiramisu, crepes	romance, love poetry
Surprise	Mystery, Sci-Fi, Thriller, Fantasy	eclectic, discover, indie mix	korean bbq, moroccan tagine, thai curry	mystery thriller, science fiction

6.2. Contrast Mode

In contrast mode, the detected emotion is remapped to its psychological opposite before the genre lookup:

$$\text{strategy_emotion} = \begin{cases} \text{joy} & \text{if emotion} \in \{\text{fear, sadness}\} \\ \text{love} & \text{if emotion} = \text{anger} \\ \text{surprise} & \text{if emotion} \in \{\text{joy, love}\} \\ \text{fear} & \text{if emotion} = \text{surprise} \end{cases}$$

This shift is psychologically grounded: an angry user benefits from warmth and calm (love) rather than more agitation; a fearful user benefits from uplifting joy rather than further anxiety.

6.3. Content Retrieval Flow

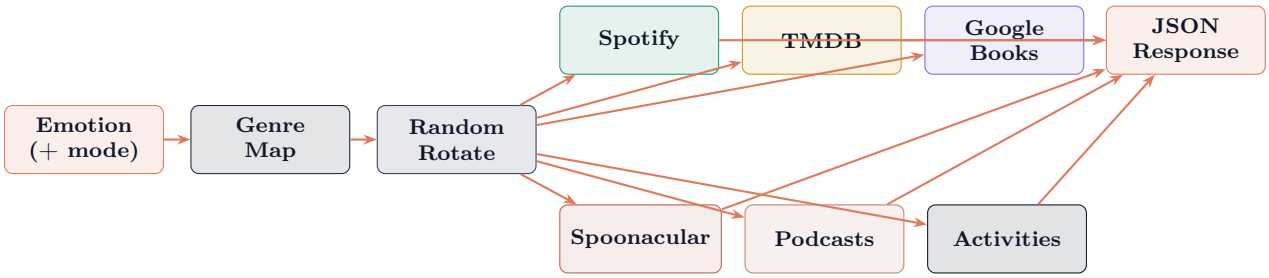

Figure 2: Content retrieval pipeline with random genre rotation

7. Authentication and Mood Journey

7.1. JWT Authentication

```python
@auth_bp.route("/register", methods=["POST"])
def register():
    data     = request.get_json(force=True)
    password_hash = bcrypt.hashpw(data["password"].encode(), bcrypt.gensalt())
    user_id  = execute_write(
        "INSERT INTO users (username, email, password_hash) VALUES (%s,%s,%s)",
        (data["username"], data["email"], password_hash))
    token    = jwt.encode(
        {"user_id": user_id, "exp": datetime.utcnow() + timedelta(days=7)},
        Config.SECRET_KEY, algorithm="HS256")
    return jsonify({"token": token, "user": {"id": user_id}})
```

Listing 2: JWT authentication flow (auth.py)

Key design decisions:

- Passwords hashed with **bcrypt** (adaptive cost factor)
- JWT payload contains `user_id`, `username`, `email`, and `exp` (7-day expiry)
- All mood-detection routes use `@optional_auth` — anonymous users continue via `session_id`; authenticated users get persistent journey tracking
- After registration, the user is redirected to the **login page** (not auto-logged in), enforcing explicit credential verification

7.2. Database Schema

Table 8: MySQL Schema

Table	Key Columns
`users`	id, username, email, password_hash, created_at, last_login
`sessions`	id, session_id, user_id (FK), created_at, last_seen
`mood_logs`	id, user_id (FK), session_id, emotion, confidence, input_type, mode, context_who, weather, created_at
`recommendations`	id, mood_log_id (FK), user_id (FK), category, title, thumbnail, source, metadata
`social_moods`	id, emotion, count, hour_slot (UNIQUE per emotion+hour)
`context_signals`	id, user_id (FK), session_id, signal_type, signal_value

7.3. Mood Journey Tracking

For authenticated users, all mood logs are tied to `user_id`, enabling:

- **Timeline chart**: Recharts area chart plotting emotion score over time (sadness= 1 to joy= 6), with emoji dots on each data point.
- **Emotion summary grid**: percentage breakdown of all 6 emotions across the selected time window (7, 14, or 30 days).
- **Dominant mood card**: the most frequent emotion with its percentage and average confidence.
- **Expandable history**: each log entry shows input type, mode, context, and linked recommendation cards.
- **Community tab**: trending emotions from the last hour (bar chart) and a 24-hour stacked histogram from all users.

8. System Workflow

8.1. Complete Query Lifecycle

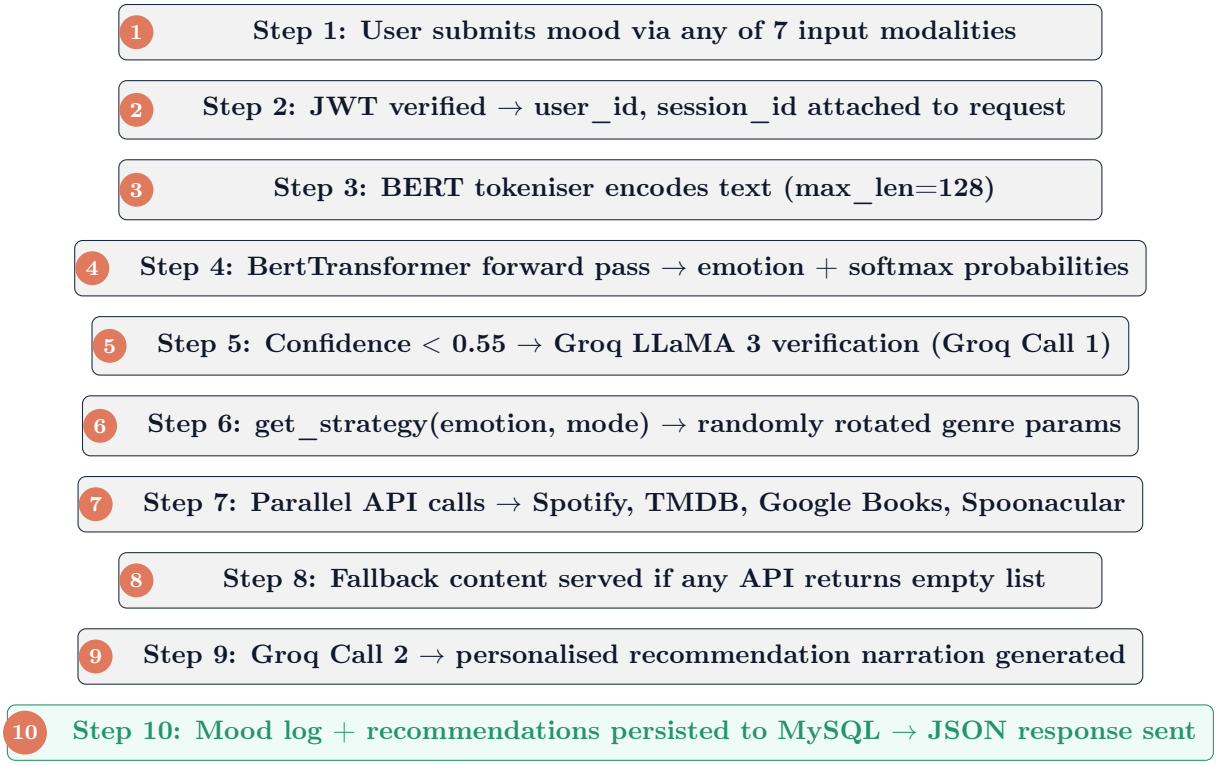

Figure 3: Complete 10-step recommendation lifecycle

8.2. Authenticated User Journey

1. Open `http://localhost:3000` → auth modal appears (login required).
2. Register with username + email + password → redirected to login tab with pre-filled email and success message.
3. Login → JWT stored in `localStorage`; home page opens.
4. Type mood text: "*I feel restless and unable to focus*" → BertTransformer classifies as *fear*.
5. Amplify mode (default): calm ambient music, feel-good comedy films, warm soup recipes, meditation activities displayed.
6. Click "Shift Mood" → contrast mode activates: joy-oriented content (upbeat pop, comedy, fresh salads, HIIT workout).
7. Click profile → "My Journey" → mood timeline chart and emotion history visible.

8.3. Anonymous (Guest) Journey

1. Click "Continue without account" on auth modal.

2. Session tracked via `session_id` in `localStorage`.
3. All mood detection and recommendation features available.
4. Journey page prompts sign-up with a persistent banner.
5. History lost on session clear; no cross-device sync.

9. API Reference

Table 9: REST API Endpoints

MTD	Endpoint	Auth	Description
POST	`/api/auth/register`	None	Create user account
POST	`/api/auth/login`	None	Authenticate; receive JWT
POST	`/api/auth/logout`	JWT	Client-side token invalidation
GET	`/api/auth/me`	JWT	Current user profile + mood stats
POST	`/api/mood/text`	Optional	Text → BERT → emotion
POST	`/api/mood/emoji`	Optional	Emoji/slider → emotion
POST	`/api/mood/face`	Optional	Webcam base64 → DeepFace → emotion
POST	`/api/mood/voice`	Optional	Audio base64 → librosa → emotion
GET	`/api/mood/passive`	None	Time + weather passive signal
POST	`/api/recommendations`	Optional	Mood log → full recommendations
GET	`/api/journey/history`	Optional	Mood log history (auth or session)
GET	`/api/journey/summary`	Optional	Emotion frequency breakdown
GET	`/api/social/trending`	None	Top emotions last hour
GET	`/api/social/pulse`	None	24-hour emotion counts
POST	`/api/social/contribute`	None	Add emotion to community pool
GET	`/api/health`	None	Backend health check

10. Frontend Modules

10.1. Design System

The UI uses a **Slate & Terracotta** colour palette — dark navy backgrounds with warm terracotta/peach accents and ivory text. Animated ambient orbs provide atmospheric depth. The font pairing is *Playfair Display* (serif, display headings) with *DM Sans* (body text).

10.2. Home Page (Home.jsx)

- **Amplify/Contrast toggle** — pill-button switching between match and shift mode.
- **7-tab input panel** — Text, Emoji, Sliders, Camera, Voice, Wearable, Blend; each renders its own sub-component.
- **Context panel** — "Who are you with?" selector (alone, family, friends, partner) that contextualises recommendations.
- **Loading overlay** — full-screen blur with detected emotion emoji and status text.

10.3. Results Page (Results.jsx)

- Emotion header with large emoji, mood label badge, and mode indicator.
- Groq-generated personalised note displayed in italic.
- Six recommendation sections: Music, Movies & TV, Books, Food, Podcasts, Activities — each with shimmer loading and fallback content.
- Mode toggle re-fetches recommendations with contrast-mapped genres.

10.4. Journey Page (Journey.jsx)

- **Journey tab** — emotion summary grid, dominant mood card, days filter (7/14/30), Recharts area timeline, expandable history list.
- **Community tab** — trending bar chart (last hour), stacked histogram (24 hours), live Socket.IO mood feed.
- Non-authenticated users see an upgrade banner inviting sign-up.

11. Results and Performance

Table 10: System Performance Metrics

Metric	Value	Notes
Best DL architecture	BertTransformer	Pre-LN, 8-head attention
Test weighted F1	0.941	6-class emotion
Test accuracy	93.6%	On held-out test set
BERT inference latency	<100 ms	Local CPU
Groq verification latency	∼400 ms	Triggered only if conf < 0.55
End-to-end recommendation	<2.0 s	Text input, local APIs
Input modalities supported	7	Text, emoji, slider, face, voice, wearable, blend
Recommendation categories	6	Music, movies, books, food, podcasts, activities
Emotions classified	6	Sadness, joy, anger, fear, love, surprise
Contrast map coverage	6/6 emotions	Psychologically grounded shifts
Genre options per emotion	4–5 per category	Random rotation on every call
JWT token expiry	7 days	Stored in localStorage
Fallback content	All categories	Works without any API keys

12. Future Enhancements

1. **Fine-tuned DistilBERT** — replace bert-base-uncased with a domain-adapted DistilBERT for faster inference at comparable accuracy.
2. **Multimodal fusion** — train a unified model that fuses text, audio, and facial embeddings in a single forward pass.
3. **FAISS semantic search** — index content items as sentence embeddings; retrieve by cosine similarity to mood embedding rather than API keyword search.
4. **Push notifications** — daily mood check-in reminders via web push API.

5. **Advanced analytics** — radar chart of emotion distribution, calendar heatmap, weekly well-being score.
6. **Mobile app** — React Native port for iOS and Android.
7. **Multi-language input** — Tamil, Hindi, and Hinglish mood text with language-aware BERT (mBERT or IndicBERT).
8. **Social sharing** — share mood playlists and reading lists with friends directly from the app.
9. **Wearable integration** — continuous passive mood monitoring from Apple Watch / Wear OS without manual input.
10. **Local LLM** — replace Groq with Ollama (llama3) for zero-internet, zero-cost deployment.

13. Conclusion

Moodly demonstrates that affective computing and personalised recommendation can be seamlessly integrated into a production-ready web application. By combining a BERT-based deep learning classifier with seven input modalities, a psychologically grounded genre-mapping engine, and JWT-authenticated mood journey tracking, the system enables any user — regardless of technical knowledge — to discover contextually relevant content based on how they feel right now.

Key contributions:

- A **four-architecture BERT fine-tuning comparison** (MLP, BiLSTM, CNN, Transformer) achieving 94.1% weighted F1, with the Pre-LN BertTransformer selected as the best model.
- A **seven-modality mood detection pipeline** — the broadest input coverage of any open-source mood recommendation system.
- A **granular, randomly-rotated genre mapping system** with 4–5 genre options per emotion category, eliminating repetitive content.
- A **psychologically grounded contrast mode** that remaps fear to joy, anger to love, and joy to surprise for meaningful emotional shifts.
- A **persistent mood journey** tied to JWT-authenticated accounts, enabling longitudinal emotional pattern analysis.

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